Radar model card, derived resolution budget, and public-proxy boundaries

Branch	Band	BW MHz	ΔR m	PRF / CRF Hz	CPI ms	Pulses/chirps	ΔfD Hz	Δv m/s
High-resolution X/Ku C-UAS	X/Ku	600	0.250	4000/4000	6.0	24	166.7	2.498
Tactical S-band AESA/MHR	S	180	0.833	4000/4000	6.0	24	166.7	8.059
GBAD 3D/4D cueing	X/Ku public cueing e	300	0.500	4000/4000	6.0	24	166.7	2.426

Public-proxy assumptions

- positive class: fixed-wing pusher-prop public proxy
- geometry: delta / swept fixed wing, rear pusher propulsor
- speed envelope: 45--60 m/s proxy
- phase windows: take-up 0--30 s, climb 30--90 s, cruise 90--150 s
- launch proxy: rail/catapult take-up with short booster assist
- RCS/aspect envelope: -28 to -2 dBsm public-source proxy
- prop micro-Doppler proxy: 150--217 Hz

Receiver impairments

- AGC compression
- clock drift
- quantization
- dropped CPI

Cue definitions

- passive RF: no-signal / RFI burst / clock-offset / provenance-quality geometry
- acoustic: node-wise spectral cadence, cross-node agreement, and amplitude stability

All values are public-proxy or synthetic archetype assumptions; no measured-platform truth is implied.